Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/21/2009
PI Name:	Ianzini, Fiorenza Ph.D.		
Project Title:	Role of high-LET radiation-induced mitotic catastrophe in mutagenesis: implication for carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	fiorenza-ianzini@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319 335 6637
Organization Name:	University of Iowa		
PI Address 1:	Pathology 145B MRC		
PI Address 2:	200 Hawkins Dr		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1009	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	12/01/2006	**End Date:**	05/31/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE until 5/31/2011 per PI and C. Guidry/JSC (02/03/2011) NOTE: End date changed to 11/30/2010 per S. Krenek/JSC (7/2010) NOTE: End date change to 9/30/2010 (from 11/30/2009) per JSC info (12/2009)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Mackey, Michael (University of Iowa)		
Grant/Contract No.:	NNJ06HH68G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This Project aims at testing the hypotheses that iron and proton ions induce mutations in exposed human cells via processes occurring in cells that undergo radiation-induced mitotic catastrophe (MC) (reviewed in Ianzini and Mackey 2007) and that this induced instability correlates with increased cancer risk. During the first 2 years of funding we have focused our efforts on determining the rate of cell killing, mutation induction, expression of regulators of mitosis (namely cyclin B1 and its phosphorylated form), production and persistence of DNA damage, and event analyses in live cell imaging by using the Large Scale Digital Cell Analysis System (LSDCAS) (Ianzini and Mackey 2002). We have reported the results pertinent to those measurements in the previous Task Books. At this time we are using our already irradiated and frozen samples to complete the analysis of DNA damage production and mutation induction after iron and proton ion exposures to increase the statistical power of our data. During the past year we have made new exiting observations and these are discussed in the Task Progress below.
References: Ianzini F. and Mackey M.A. 2007. Mitotic Catastrophe. In “Apoptosis, Senescence and Cancer”, Second Edition (D. A. Gewirtz, S. E. Holt, S. Grant eds.), Humana Press; pp. 73-91, 2007.

Ianzini F. and Mackey M.A. 2002. Development of the Large-Scale Digital Cell Analysis System. Radiat Prot Dos 99, 81-94.

Rationale for HRP Directed Research:

The studies conducted in this project fell in the category of risk #28 – Carcinogenesis - of the Radiation Health Bioastronautics Roadmap Table. In particular, we are addressing the following questions:
28b How can tissue specific probabilities for increased carcinogenesis risk from space radiation be best evaluated? What molecular, genetic, epigenetic, abscopal (effect that irradiation of a tissue has on remote non-irradiated tissue), and other factors contribute to the tissue specificity of carcinogenic risk?

To answer this question we are testing the hypothesis that high-LET radiation leads to an enhanced incidence of MC (definition and description of MC is reported in the “Task Description” section). The idea is to define MC susceptibility of human cells exposed to densely ionizing radiation to be used as a “biomarker” for delayed effects of exposure to high-LET radiations

28i What are the probabilities that space radiation will produce DNA damage at specific sites, including clustered DNA damage? What is the likelihood that DNA damage will increase the risk of carcinogenesis?

Research Impact/Earth Benefits:

To answer this question we are testing the hypothesis that DNA damage is persistent in survivors of cells that have undergone radiation-induced-MC and that mutation induction is elevated in these survivors.

28j What mechanisms modulate radiation damage at the molecular level (e.g., repair, errors in repair, signal transduction, gene amplification, bystander effects, tissue microenvironment, etc.) that significantly impact the risk of cancers, modulate the expression of radiation damage and decrease the radiation risk, and how can the understanding of mechanisms be used to predict carcinogenic risks from space radiation?

To answer this question we are following the irradiated cell populations for long time post-irradiation by means of live cell imaging. Our previous observation using the Large Scale Digital Cell Analysis System (LSDCAS) – an in-house built live cell imaging system extremely versatile – has shown that a small proportion of irradiated cells undergoing MC escape death. We are following these survivors to determine their viability. If these cells form viable colonies and are persistent for long time the risk of cancer development under these conditions can be deemed increased.

Task Progress:

One of the most important findings of our studies is that we were able to demonstrate that all the cell lines we have analyzed (murine or human, embryonic, fetal or adult) undergo radiation-induced MC and that they are able to survive for weeks post-irradiation. These results, coupled with findings on cancer cells (Ianzini et al. 2009) that demonstrated that irradiated cells undergo MC, become polyploid, and are able to escape death (at least for a month post-irradiation) by activating processes of depolyploidization that utilize uncharacteristic pathways of divisions (namely activation of meiotic genes during multipolar divisions), prompted us to hypothesize that similar processes might occur in normal cells exposed to space radiation.
The potential occurrence of these processes, associated with the production of persistent DNA damage, is of great importance as it would imply that normal cells that receive a dose of space radiation and undergo MC can escape death by re-organizing their genomic composition. If these cells are also able to proliferate (as we have demonstrate being the case for a small percentage of cancer cells that undergo depolyploidization – Ianzini et al. 2009), their progeny would likely have a disarrayed DNA that could be prone to further mutations and ultimately render this cell progeny more susceptible to carcinogenesis. Believing that this reasoning would broaden our understanding of specific pathways of survival in irradiated normal cells we devoted our last year funding efforts towards demonstrating that pathways of meiotic division are elicited in normal cells that escape death after having been exposed to space radiation. In deed we were able to demonstrate that the meiosis-specific SPO11 and SCP3 genes are expressed in the normal cells we have analyzed along with nestin, a specific markers of cancer stem cells.

These findings support the hypothesis that normal cells that have undergone radiation-induced MC, but escaped death, may retain proliferative abilities facilitated by the activation of unusual pathways of cell division that utilize meiosis-specific proteins and that the progeny thus originated acquires genetic and/or phenotypic characteristics of cancer stem cells. Thus, it is possible that aberrant expression of stem cell genes may occur via the activation of silenced gametogenic programs in cells exposed to charged particle radiation and that this switch may play a role in carcinogenesis.

Finally, we also used some of the irradiation time that was allotted to us to acquire preliminary data for a second project within NASA interests (effects of space radiation on central nervous system (CNS)). This project is a collaborative effort among the University of Iowa, Loma Linda University and the University of California at San Francisco and it aims at testing the hypothesis that neural stem cells (NSC) exposed to different energies of proton and silicon ions, and proton-iron mixed field radiations are functionally altered, effectively resulting in decrements of their ability to migrate, proliferate, integrate and reduce injury within the brain of an animal model of trauma. Within this project we have gained preliminary data on DNA damage and motility characteristics of NSC irradiated with proton and iron ions. In particular, in flowcytometry and immunofluorescence (IF) measurements, we were able to demonstrate that NSC irradiated with a

	dose of 0.5 Gy of 40 MeV protons retain proliferative capacity throughout the course of the experiment (10 days), and that parameters of cell cycle remain unaltered during the course of the experiment. Thus, this dose is non-perturbing to the NSC. Measurements of DNA damage (IF – gamma-H2AX expression) showed that NSC are in general able to repair radiation-induced DNA damage, with iron ions being more effective than protons at producing DNA damage. Finally, LSDCAS live cell imaging results demonstrated that gamma-, 40 MeV proton, and 1 GeV iron ion radiations have minimal effect on cell growth and cell death on NSC. However, iron ion radiation induced an increase of about 6 h in the NSC doubling time, slowing cell division. Thus, it appears that iron ions have a detrimental effect on proliferation in NSC. This effect is confirmed by the LSDCAS motility experiments in which it is showed that iron ion radiation depress NSC motility. Thus, taken together, this in vitro characterization of the NSC by cell cycle, molecular, and imaging analyses, reveals that these cells are moderately affected by the relatively low dose of proton irradiation, but cell division and cell motility are more affected after iron ion exposure. These preliminary results pave the way towards understanding how space radiation might impair neural stem cells competence. References Ianzini F., Kosmacek E.A., Nelson E.S., Napoli E., Erenpreisa J., Kalejs M., Mackey M.A. 2009. Activation of meiosis-specific genes is associated with depolyploidization of human tumor cells following radiation-induced mitotic catastrophe. Cancer Res., 69, 2296-2304.
Bibliography Type:	Description: (Last Updated: 02/12/2021)
Abstracts for Journals and Proceedings	Kosmacek EA, Napoli E, Szyperski MR, Keck KJ, Schwertner AB, Mackey MA, Ianzini F. "Cell Cycle Dysregulation and Mitotic Catastrophe After Exposure of Murine and Human Cell Lines to 1 GeV Iron Ion Irradiation." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, USA, June 30-July 2, 2008. Abstract book. 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, USA, June 30-July 2, 2008. , Jun-2008
Abstracts for Journals and Proceedings	Ianzini F, Kosmacek EA, Napoli E, Szyperski MR, Schwertner AB, Bryant AM, Mackey MA. "Role of High-LET Radiation-Induced Mitotic Catastrophe in Carcinogenesis." NASA Human Research Program Investigators' Workshop, League City, TX, USA, February 2-4, 2009. Abstract book. NASA Human Research Program Investigators' Workshop, League City, TX, USA, February 2-4, 2009. , Feb-2009
Abstracts for Journals and Proceedings	Ianzini F, Napoli E, Kosmacek EA, Mackey MA, Ashwal S, Snyder E, Obenaus A. "Functional Alterations in Neural Stem Cells Following Space Relevant Radiation Exposures." Heavy Ion Symposium and 20th NASA Space Radiation Investigators Workshop, Cologne, Germany, July 6-10, 2009. Abstract book. Heavy Ion Symposium and 20th NASA Space Radiation Investigators Workshop, Cologne, Germany, July 6-10, 2009. , Jul-2009
Abstracts for Journals and Proceedings	Ianzini F, Napoli E, Kosmacek EA, Szyperski MR, Schwertner AB, Mackey MA. "MRC-5 Human Lung Fibroblast Cells Exposed to 1 GeV 56Fe Ions Express the Meiosis-Specific Proteins SPO11 and SYCP3 and the Cancer Stem Cell Protein Marker Nestin." Heavy Ion Symposium and 20th NASA Space Radiation Investigators Workshop, Cologne, Germany, July 6-10, 2009. Abstract book. Heavy Ion Symposium and 20th NASA Space Radiation Investigators Workshop, Cologne, Germany, July 6-10, 2009. , Jul-2009
Abstracts for Journals and Proceedings	Obenaus A, Ashwal S, Snyder E, Napoli E, Kosmacek EA, Mackey MA, Ianzini F. "Alterations in Neural Stem Cells Following Ionizing Radiation Exposures: Functional Magnetic Resonance Imaging and Live Cell Imaging Studies." 55th Annual Meeting of the Radiation Research Society, Savannah, GA, USA, October 4-7, 2009. Abstract book. 55th Annual Meeting of the Radiation Research Society, Savannah, GA, USA, October 4-7, 2009. , Oct-2009
Abstracts for Journals and Proceedings	Ianzini F, Kosmacek EA, Napoli E, Szyperski MR, Schwertner AB, Mackey MA. "Tumor and Normal Human Cell Lines Exposed to Sparsely and Densely Ionizing Radiation Express Meiotic-Specific Proteins and Cancer Stem Cell Markers." 55th Annual Meeting of the Radiation Research Society, Savannah, GA, USA, October 4-7, 2009. Abstract book. 55th Annual Meeting of the Radiation Research Society, Savannah, GA, USA, October 4-7, 2009. , Oct-2009
Articles in Peer-reviewed Journals	Ianzini F, Kosmacek EA, Nelson ES, Napoli E, Erenpreisa J, Kalejs M, Mackey MA. "Activation of meiosis-specific genes is associated with depolyploidization of human tumor cells following radiation-induced mitotic catastrophe." Cancer Res. 2009 Mar 15;69(6):2296-304. PubMed PMID: 19258501 , Mar-2009
Articles in Peer-reviewed Journals	Sun Y, Davis PJ, Kosmacek EA, Ianzini F, Mackey MA. "An open-source deconvolution software package for 3-D quantitative fluorescence microscopy imaging." Journal of Microscopy 2009 August, Early View Published Online: 25 Aug 2009. http://dx.doi.org/10.1111/j.1365-2818.2009.03205.x , Aug-2009